

BANKHAUL · DESIGN DECISIONS REGISTER

BankHaul design decisions register

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	BKH-DEC-001	Version	0.1
Status	Draft	Date	2026-10-03
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/bankhaul		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	Register opened; design decisions made under Amish's pre-approvals of 2026-10-03; two requirement decisions proposed

BankHaul design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` or in `docs/REVIEW.md`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list decisions.

Safety: Several decisions below set BankHaul's safety case (weak links, working distance, stayed post, far stake, floods). Each took the conservative option; the evidence that would relax it is in BKH-DDR-001, Table 1.

■ Open decisions

Two requirements are not met or at risk on paper. They are not decided under the pre-approval; the state, options and recommendation for each are set out in `docs/REVIEW.md`, TRL 3 section, "Decisions for Amish".

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	R5, far stake holding in very soft mud: factor 1.23 on the 1.5 kN load where the bed's shear strength is about 5 kPa	A: drive 3.5 m into very soft beds (5.0 m stake); B: two stakes 1 m apart, tied at the top; C: keep 2.5 m and use only sites with a firmer bed	Proposed, awaiting Amish. Recommend A	Stake length and mass (section 3.2, step 6)	BKH-CAL-001, E; <code>docs/REVIEW.md</code>
2	R8, cost per station: USD 382	A: build the steel station for the TRL 4 trials and cost a local-materials station with the co-design partner; B: switch the prototype to local materials now (about USD 98); C: one bank post and table for three loops (about USD 257 per loop)	Proposed, awaiting Amish. Recommend A	None if A; most made parts if B; the post head if C	BKH-CAL-001, I; <code>docs/REVIEW.md</code>

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The weak link cord: three samples from the batch each release between 500 and 700 N as a single 80 mm loop	Sets the largest load a seized net can put into the system (R10)	BKH-CAL-001, C
2	The screw anchors' helix diameter, rod length and eye, and the supplier's holding figure in soft clay	The bank check uses a 250 mm helix 0.9 m deep (R4)	BKH-CAL-001, D
3	The blocks' eye and cheek-spacer positions sit outside the rope's path round the sheave, and the sheave takes 12 mm rope	The loop must not rub the block; the post and collar holes are set for this block length	BKH-DDR-002, item 9
4	The jam cleats' fixing hole spacing	The cleat bar is drilled for 60 mm	BKH-DWG-101
5	The rope's breaking strength, floating and UV stabilisation	Factor 5.7 after a season assumes 20 kN new and half lost to sun	BKH-CAL-001, F
6	The turnbuckles and shackles are rated and marked with their working load limits	They carry the stays at up to 2.1 kN	BKH-CAL-001, D
7	The bed's shear strength at the trial site (a hand vane or cone test from the boat)	Decides whether open decision 1 applies there	BKH-CAL-001, E

■ Value engineering

Value-engineering target: USD 1,800 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 392 including the stake driving cap (USD 1,408 under the target). Main cost drivers and savings worth trying:

- The two screw anchors (USD 76) are the largest line, then the rope (USD 54), the far stake (USD 48), the two blocks (USD 44), the table (USD 35) and the post (USD 32).
- Rated rigging (shackles, turnbuckles, blocks) is kept because it carries the safety case.
- Savings worth trying: rope bought only for the site's span (USD 29 for 30 m); a local timber table; anchors bought in bulk through a fencing or utility supplier; a hardwood far pole where the bed is firm.

■ Decisions made

The pre-approvals: Amish, 2026-10-03: "start with the first 14 repos from the list of 29 projects. I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost." and, for this batch, "Proceed with the remaining 15 scaffolds".

Date	Decision	Decided by	Record
2026-10-03	Bank anchor: post on a foot plate, held back by two stays to two screw anchors	Amish, under both pre-approvals quoted above	BKH-DDR-001, item 1
2026-10-03	Driven stake for the first prototype; float kept as a variant for deep beds	Amish, under both pre-approvals quoted above	BKH-DDR-001, item 2
2026-10-03	Weak link at each net end, 500 to 700 N; requirement R10 added	Amish, under both pre-approvals quoted above	BKH-DDR-001, item 3
2026-10-03	Fisher works at least 5 m from the water; post 6 m back; requirement R11 added	Amish, under both pre-approvals quoted above	BKH-DDR-001, item 4
2026-10-03	12 mm floating polysteel loop	Amish, under both pre-approvals quoted above	BKH-DDR-001, item 5
2026-10-03	Two swivel rings half a loop apart as clip rings and end stops; net at least 2 m shorter than the span	Amish, under both pre-approvals quoted above	BKH-DDR-001, item 6
2026-10-03	A jam cleat for each leg as the bank brake	Amish, under both pre-approvals quoted above	BKH-DDR-001, item 7
2026-10-03	Haul by the net's near end with the loop following	Amish, under both pre-approvals quoted above	BKH-DDR-001, item 8
2026-10-03	Loop taken out before floods	Amish, under both pre-approvals quoted above	BKH-DDR-001, item 9
2026-10-03	First co-design candidate to approach: the Lake Kariba research group of the Oryx study with an inshore fishing cooperative (not agreed)	Amish, under both pre-approvals quoted above	BKH-DDR-001, item 10
2026-10-03	CalRig as the first candidate proof-load rig; no hand capstan	Amish, under both pre-approvals quoted above	BKH-DDR-001, item 11
2026-10-03	budget_usd kept at 1,800 as a value-engineering target	Amish: "I also accept any cost overruns or variations from the assumed scope cost."	BKH-DDR-001, item 12
2026-10-03	Design for construction: the twelve changes of BKH-DDR-002	Amish, under both pre-approvals quoted above	BKH-DDR-002
2026-10-03	Appearance model additions for renders: loop cut short, stake above the bed only, net on the table, mannequin beside the post	Amish, under both pre-approvals quoted above	docs/REVIEW.md , TRL 3